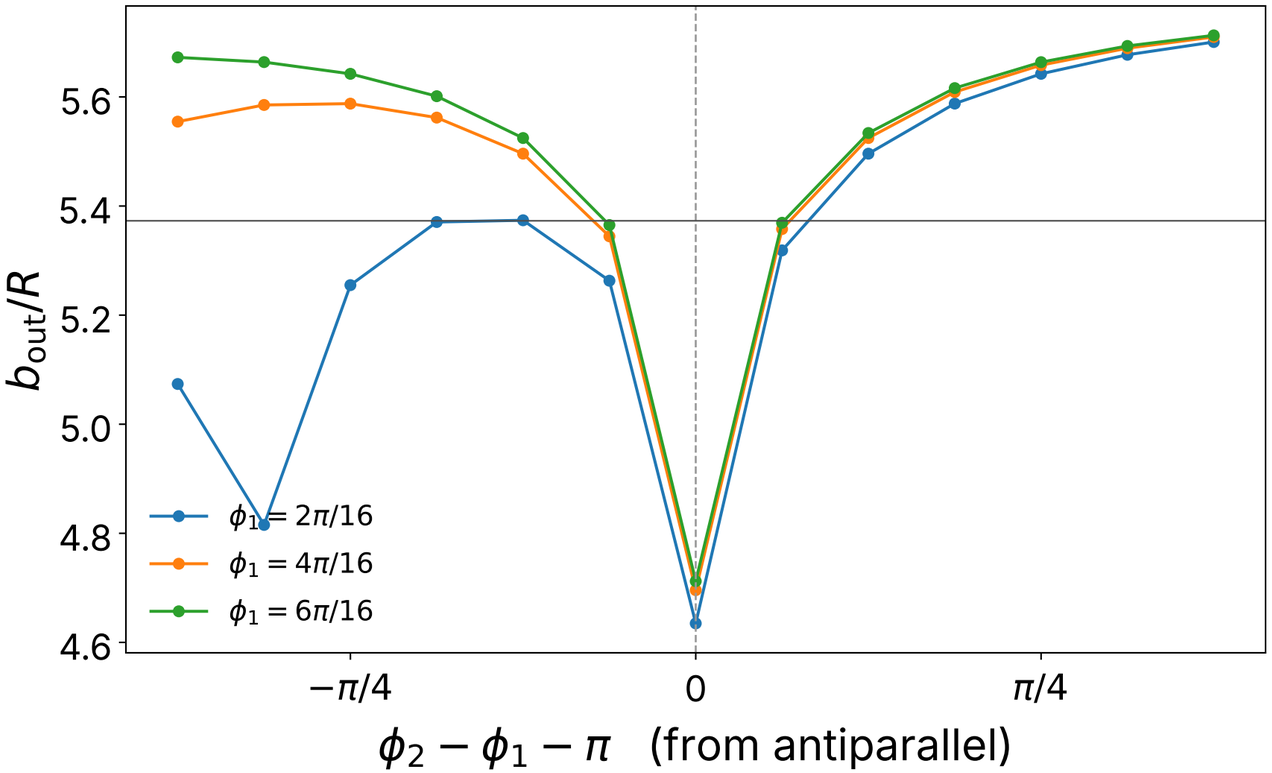
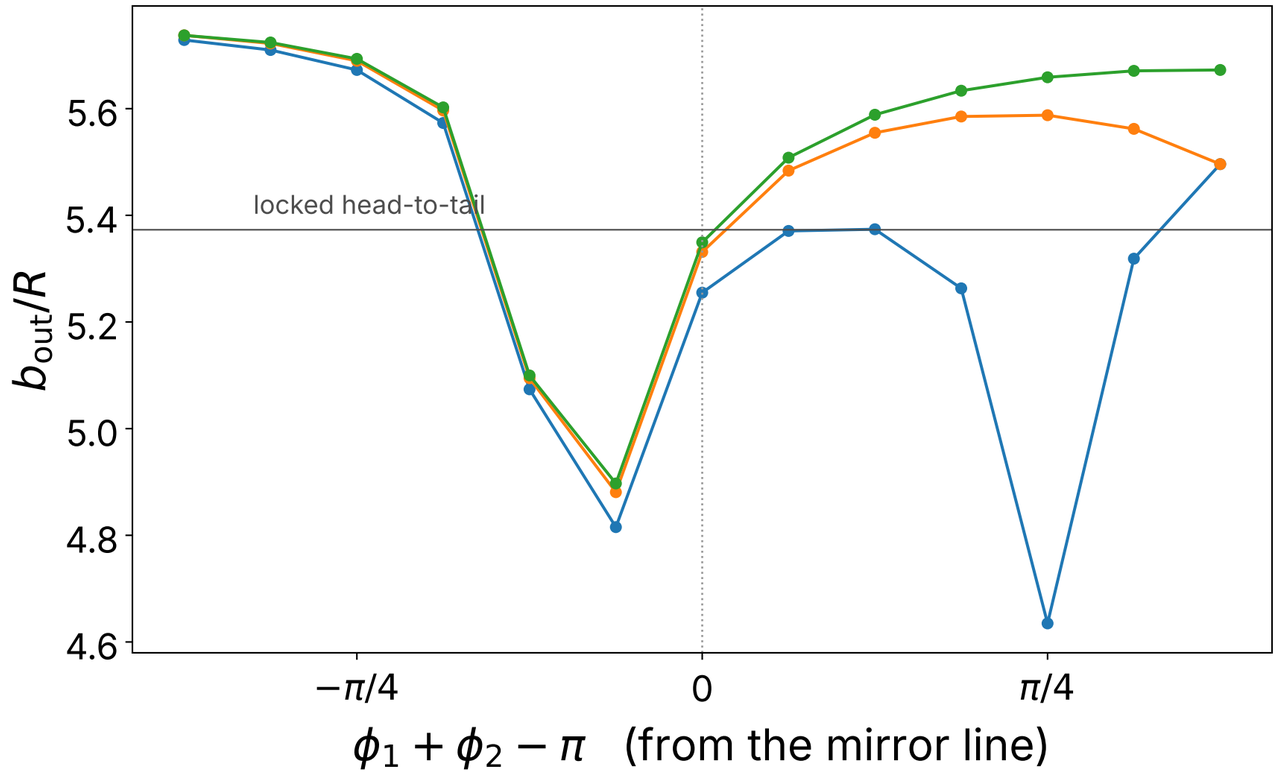
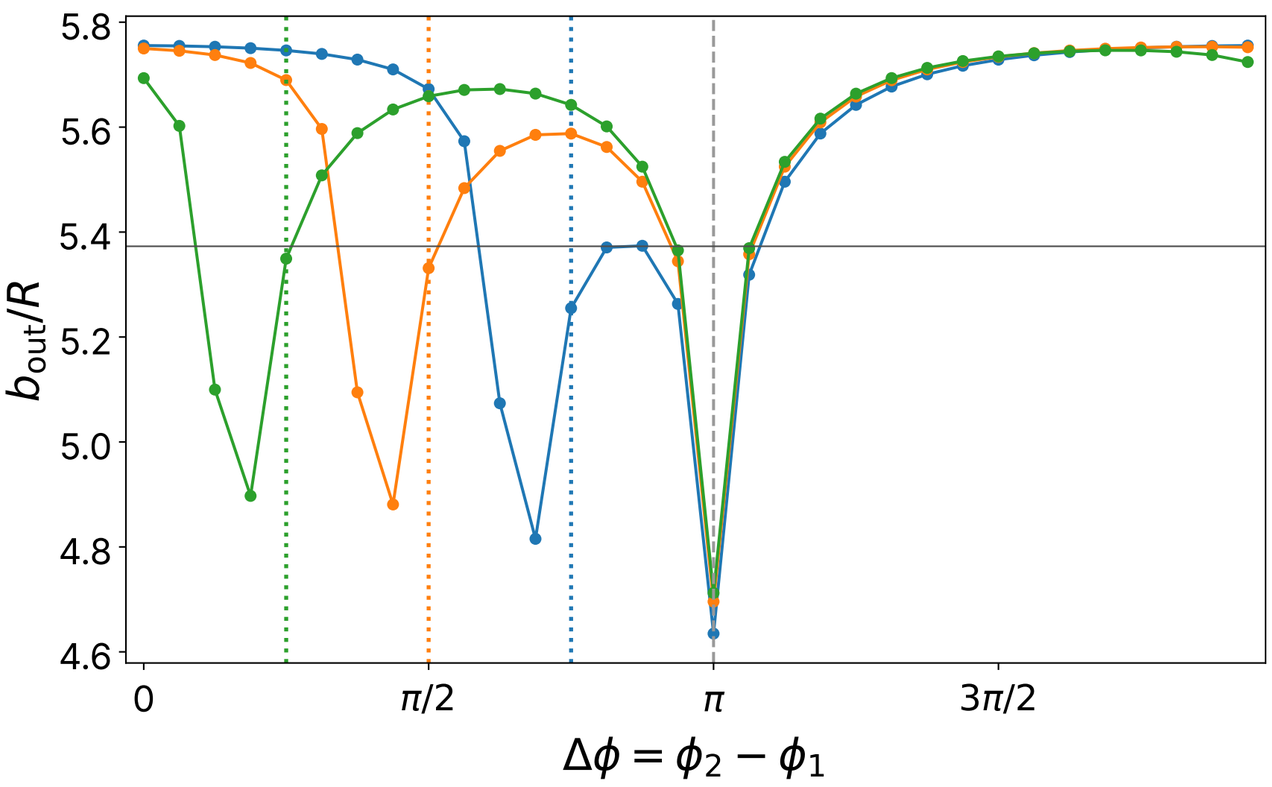
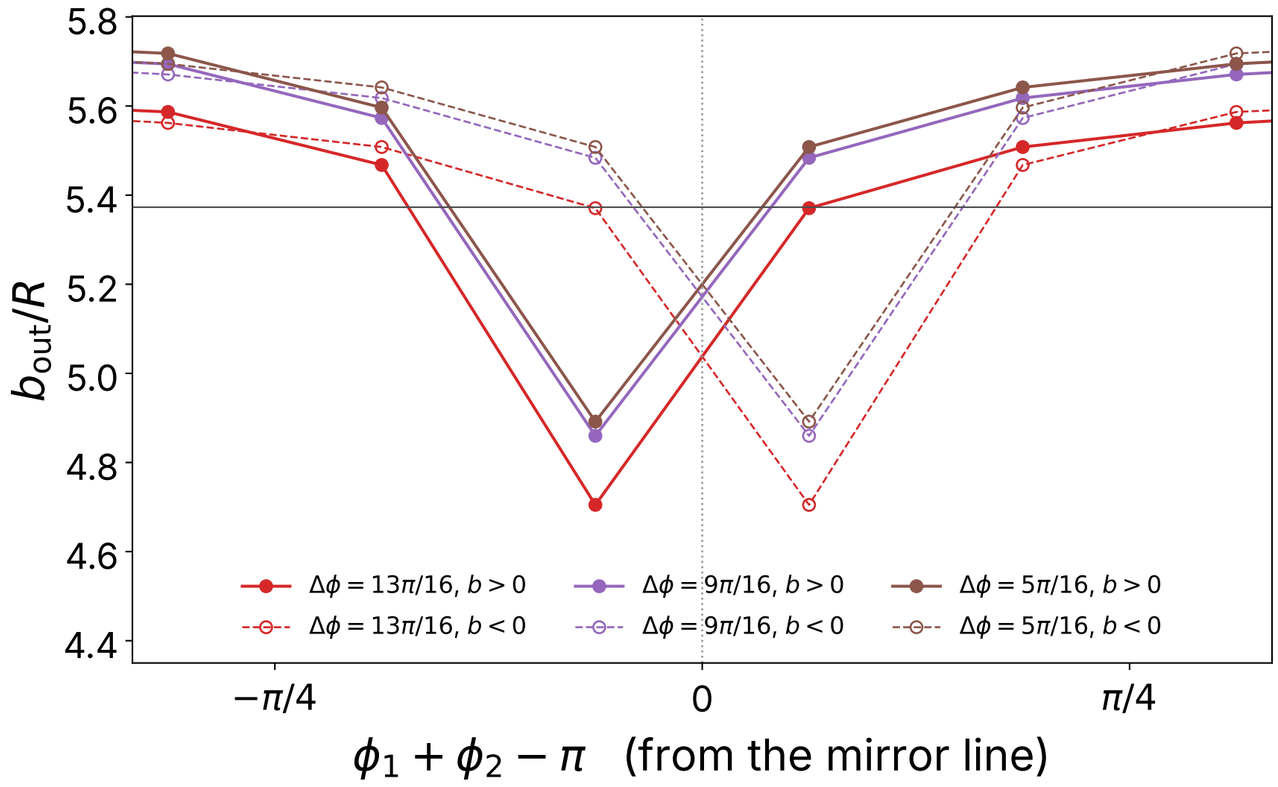

b_{out}/R
5.6
5.4
5.2
5.0
4.8
4.6
$\phi_1 = 2\pi/16$
$\phi_1 = 4\pi/16$
$\phi_1 = 6\pi/16$
$-\pi/4$
0
$\pi/4$
$\phi_2 - \phi_1 - \pi$ (from antiparallel)


b_{out}/R
5.6
5.4
5.2
5.0
4.8
4.6
locked head-to-tail
$-\pi/4$
0
$\pi/4$
$\phi_1 + \phi_2 - \pi$ (from the mirror line)


b_{out}/R
5.8
5.6
5.4
5.2
5.0
4.8
4.6
0
$\pi/2$
π
$3\pi/2$
$\Delta\phi = \phi_2 - \phi_1$


b_{out}/R
5.8
5.6
5.4
5.2
5.0
4.8
4.6
4.4
$\Delta\phi = 13\pi/16,\ b > 0$
$\Delta\phi = 13\pi/16,\ b < 0$
$\Delta\phi = 9\pi/16,\ b > 0$
$\Delta\phi = 9\pi/16,\ b < 0$
$\Delta\phi = 5\pi/16,\ b > 0$
$\Delta\phi = 5\pi/16,\ b < 0$
$-\pi/4$
0
$\pi/4$
$\phi_1 + \phi_2 - \pi$ (from the mirror line)